National Computer Education Accreditation Council NCEAC

NCEAC.FORM.001-C

INSTITUTION National University of Computer & Emerging Sciences, Islamabad

PROGRAM (S) TO BE EVALUATED Computer Science (BS) – Spring, 2024

Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	CS2005
Course Title	Database Systems
Credit Hours	3 Classes and 1 Credit Lab for 3 hours (3-1-3)
Prerequisites by Course(s) and Topics	Data Structures (CS2001)
Grading Policy	Absolute as per NUCES grading policy
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pre-take cases.
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course as per university policy. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	100% theory and 1 credit separate lab Breakdown Assessment Items

Assessment Item	**Number**	**Weight (%)**
Quizzes	>=4	10
Assignments/HWs	3	10
Project	1	10
Sessional-I Exam	1	15
Sessional-II Exam	1	15
Final Exam	1	40

Course Coordinator	Ms. Noor ul Ain
Course Instructors	Dr. Syed Qaiser Ali Shah , Ms. Noor ul Ain
Lab Instructors	Ms. Amina Siddique, Mr. Hassan Ali Ansari
URL (if any)	
Current Catalog Description	Basic database concepts. Fundamentals of Information System, Relational data model. Relational theory and languages. Conceptual modeling & Database design. Programming

	using SQL. Introduction to transaction processing and concurrency. Introduction to query processing and optimization, database security and emerging trends.
Textbook (or **Laboratory Manual** for Laboratory Courses)	**Fundamentals of Database Systems,** by Ramez Elmasri and Shamkant Navathe, Addison Wesley, 7th Edition.
Reference Material	**Database System Concepts** by Henry F. Korth and Abraham Silberschatz, 4th edition, McGraw Hill, 2002, ISBN: 0-07-12268-0 **Database Systems: Design, Implementation, & Management** Carlos Coronel/Steven Morris, 13th Edition, ISBN: 9780357687536
Course Description & CLOs	Understanding the concepts and theory of database systems. Topics include database system architectures, data models, query languages, conceptual and logical design, physical organization, transaction management, and query processing. The entity-relationship model and relational model are discussed in detail for the basis of developing the conceptual model. The mapping from the conceptual level to the logical level, integrity constraints, functional dependencies and normalization are studied as a basis for formal design. Theoretical languages such as the relational algebra and the relational calculus are described, and high-level language, SQL, is discussed. File organization and access methods are provided as a basis for discussion of query optimization techniques. Transaction processing techniques are presented with a specific emphasis on concurrency control. Emerging approaches & technologies such as NoSQL, Data Warehousing & Business Intelligence, are also discussed. The course learning objectives are preparing students for effective use of database technology for data management with hands-on-experience in a team work environment. After completion of the course, the student shall be able to: 1. Learn fundamental concepts, architecture and components of database systems. 2. Evaluate relational model, relational algebra and constraints on data with real life examples. 3. To design a conceptual models using ER/ EER Diagram for a real life database applications. 4. Implement database schema of ER/EER Diagrams. 5. Test conceptual models or design using functional dependency and normalization. 6. Learn the concepts of database programming, SQL. 7. Learn fundamental concepts of transaction processing & concurrency. 8. Learn fundamental concepts of query processing & optimization. 9. Learn emerging approaches & technologies such as NoSQL, Big Data, Data Warehousing & Business Intelligence, Distributed Data Management.

Program Learning Outcomes

For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.

PLO 1	Computing and Cybersecurity Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solve complex problems using computer networks and cybersecurity techniques	✔
PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computational problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, computing, and cybersecurity	✔
PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✔
PLO 4	Investigation & Experimentation	Conduct investigation of complex computing and security problems using research-based knowledge and research-based methods	✔
PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing and cybersecurity tools for complex security problems	✔
PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems	
PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing and cybersecurity work in solving complex computing and security problems	
PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing and cybersecurity practice	
PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings	✔
PLO 10	Communication	Communicate effectively on complex computing and cybersecurity activities with	✔

		the cybersecurity community and with society at large	
PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team	
PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes	✔

C. Mapping of CLOs on PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes

		PLOs											
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CLOs	1	✔		✔							✔		
	2	✔		✔	✔	✔					✔		
	3	✔		✔	✔	✔					✔		
	4	✔	✔	✔	✔	✔				✔	✔		
	5	✔									✔		
	6	✔	✔	✔	✔	✔				✔	✔		
	7	✔	✔	✔	✔	✔				✔	✔		
	8	✔		✔	✔						✔		
	9	✔	✔	✔	✔						✔		✔

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-weeks instructions and one-hour lectures)

Topics to be covered:

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction to course, Introduction to Databases	**1**	**3**	**1**
Database Systems Concepts & Architecture, Data Models	**1**	**3**	**2**
Data Modeling & Entity Relationships (ER)	**2**	**6**	**3, 4**
Enhanced Entity Relationships (EER)	**1**	**3**	**3, 4**
Relational Model & Constraints	**1.66**	**4.98**	**3**
SQL Programming	**1**	**3**	**6**

Database Design, Mapping from conceptual to logical model	**1.24**	**3.72**	**3, 4**
Functional Dependencies & Normalization	**1.3**	**3.9**	**5**
Relational Algebra	**1**	**3**	**2**
Transaction Processing and concurrency control	**1.3**	**3.9**	**7**
Introduction to Query Processing & Optimization	**1**	**3**	**8**
Distributed Database Management System	**0.75**	**1.5**	**9**
Data Warehouse and BI	**0.75**	**2**	**9**
Big Data and NoSQL	**1**	**3**	**9**
Total	**16**	**47**	

Laboratory Projects/Experiments Done in the Course	Introduction to lab and database architecture. Hands on practice using SQL programming, Basic Web Programming using html and Javascripts, designing of Forms/ Reports using PHP. Basic Database Administration. Evaluation is managed with problem solving approach evaluated by lab work, practical quizzes and lab project.
Programming Assignments/ Lab Done in the Course	Microsoft SQL Server, Visual Studio, C#, MongoDB, PostgreSQL

Class Time Spent on (in credit hours, Hrs/Min)	**Theory (%)**	**Problem Analysis (%)**	**Solution Design (%)**	**Ethics (%)**
	50	25	20	5

Oral and Written Communications	Every student is required to submit at least __4__ written reports. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.

CS2005 COURSE CONTENTS (Lab/ Practical):

Weeks	Contents/Topics	**Courseware Events (MM/ IT Lab/Case Study/ Assignment/ Presentation etc.)	Comments (if any)
Week-01	Introduction to lab and database Architecture. SQL Server installation guide		
Week-02	Overview of MS Access.		
Week-03	Introduction to Web Server Architecture, Introduction to C# Programming		
Week-04	Design ER on multiple case studies	Assignment 1	
Week-05	Introduction to SQL using SQL Server (SQL-SELECT, Aggregate Functions, NVL, Usage of Operators - IN,BETWEEN, LIKE,IS NULL, AND, OR, NOT) (Lab03.docs)		
Week-06	Wild Card Search, String Functions, Views, Top-N Analysis, Working with Date and Number, Joins (Lab04.docs)	Lab Quiz-1	
Week-07	Sub-queries and DDL with constraints, DDL Script writing/ Schema Creation (Lab05.docs)	Assignment 2	
Week-08	Database connection using SQL server Getting started with Lab Project, Case Study, Defining and formulating Problem Definition. (Lab6.docs)	Case Study	
Week-09	ER mapping and script writing Introduction to HTML & Java Script (Lab07-ODBC.docs)	Assignment 3	
Week-10	C# Programming, Form Development, Reports Development using Crystal Report Application		
Week-11	Introduction to procedural programming PL/SQL (Procedures, Functions, Packages & Triggers)	Lab Project	
Week-12	Procedural programming PL/SQL (Procedures, Functions, Packages & Triggers)	Lab Quiz-2	
Week-13	Introduction to PostgreSQL, Transaction Processing and concurrency control		
Week -14	Introduction to Mongo DB		
Week 15	Project		
Week-16	Project Demo		
Week-17	Project Demo		

Practical/ Programming Work/ Tools: Microsoft SQL Server, Visual Studio, C#

Proposed assessment items of Lab Part

Assessment Item	**Number**	**Weight (%)**
Lab Tasks	10~13	20
Lab Quiz	5	10
Assignments	3~4	10
Project	1	20
Final Exam	1	40